

Intraday lottery demands in cryptocurrency market

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Abstract

Purpose – Using the high-frequency intraday data of the top 100 most liquid cryptocurrencies, this study aims to examine the presence of the MAX effect in the cross-sectional pricing of cryptocurrencies. Additionally, it delves into the pricing implications of idiosyncratic volatility (IVOL) and skewness, both idiosyncratic and systematic, across the cross-section of cryptocurrency returns and their interaction with the MAX measure.

Design/methodology/approach – Driven by the growing influence of high-frequency trading (HFT) in the cryptocurrency market, the study modifies Bali *et al.*'s (2011) MAX measure by incorporating an hourly forecast horizon and 5-min log returns from the preceding hour. The relationship between MAX, IVOL and skewness over the past hour and expected returns is examined using portfolio-level analysis and Fama and Macbeth's (1973) cross-sectional regressions.

Findings – The findings indicate that an increase of one standard deviation in MAX corresponds to a 0.043% decline in subsequent returns for cryptocurrencies, suggesting overvaluation due to increased demand. These results are robust to other traditional price determinants. Untangling the MAX from other proxies of the lottery, the study reveals that MAX is the true effect in the cryptocurrency market. The results are robust to several sensitivity checks, such as varying MAX measures and holding periods.

Originality/value – The study pioneers the investigation of lottery-like demand within cryptocurrency markets at the intraday frequency. To the best of the author's knowledge, this is the first paper untangling the association between MAX and IVOL in the cryptocurrency market and thus offers valuable insights into investor behavior in these emerging markets.

Keywords MAX effect, Idiosyncratic volatility, Lottery-like preferences, Cryptocurrency, Skewness preferences, Cross-section of returns

Paper type Research paper

1. Introduction

The meteoric rise of cryptocurrencies (CC or CCs, henceforth) in the last decade has been nothing short of a financial revolution. The mounting growth coincides with increased interest from media and finance literature. The recent spur in institutional adoption (high-profile endorsements and investments by institutional players like Tesla and PayPal) underscores the lasting popularity of digital currencies. In December 2017, the Chicago Mercantile Exchange and the Chicago Board Options Exchange initiated Bitcoin futures contract trading. Central banks are also discovering new payment tools and preparing to introduce their digital currencies (CBDCs). These developments establish them as an important asset class rather than just a medium of exchange. Literature provides several evidence of its role as a valuable asset (Bouri *et al.*, 2017; Brière *et al.*, 2015; Glaser *et al.*,



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2014). Thus, analyzing the CC market through the lens of empirical asset pricing is significant for understanding how investors can design the most efficient portfolios.

Today, the CC literature addresses almost all the standard research questions of traditional financial research, including market efficiency, bubble formation, seasonality, liquidity and volatility. The conventional factor models dominate the literature on the cross-section of CCs and have recently garnered considerable limelight (Liu *et al.*, 2019; Liu and Tsyvinski, 2021). These models suggest that since investors are rational and returns are mean-variance efficient, only systematic risk is priced (Lintner, 1965; Markowitz, 1952; Sharpe, 1964). However, several recent behavioral finance models indicate that investors frequently deviate from the models of rational equilibrium and Expected Utility Theory, often displaying a preference for lottery-like investments (Barberis *et al.*, 2008; Boyer *et al.*, 2014; Kumar, 2009).

Investors' inclination for stocks with lottery-like traits, even with them underperforming their counterparts, has garnered significant attention in the asset pricing literature (Annaert *et al.*, 2013; Bali *et al.*, 2011; Kumar, 2009; Walkshäusl, 2014). The MAX effect exemplifies this behavioral complexity, where investors paradoxically exhibit a preference for high-volatility, lottery-like assets despite their suboptimal financial performance. This preference stems from deeply ingrained cognitive biases, and the evidence gains support from several psychologically realistic models challenging the rationality assumption in the decision-making process. For instance, the probability weighting element of cumulative prospect theory (CPT) by Tversky and Kahneman (1992) states investors' tendency to overweigh the outcomes with small probabilities, reflecting their gambling and insurance preferences. In this context, retail investors tend to be influenced by past return performance, overweighing the positive minimal-probability events they observe (Barberis *et al.*, 2016; Barberis and Huang, 2008; Chelikani *et al.*, 2022) and holding on to losing instruments (Li and Yang, 2013; Meng and Weng, 2017; Odean *et al.*, 1998). This behavior also explains the investors' preferences for positive skewness even at the cost of higher expected returns (Kumar, 2009).

Additionally, theories like Brunnermeier *et al.*'s (2007) optimal expectations framework explain this preference by proposing that investors intentionally manipulate their beliefs, overvaluing the odds of outcomes with positive payoffs in their portfolio decisions. Thus, given the robust evidence of speculative behavior in equity and other conventional markets (Akbas and Genc, 2020), it is only natural to extend this examination to CC investors. This is primarily because CCs, unlike traditional assets, lack fundamental value anchors (Cheah and Fry, 2015; Mnif *et al.*, 2022), making them highly susceptible to investor sentiment and speculative behavior.

Academic interest in examining speculative behavior in CC markets has been longstanding. More specifically, concerning lottery-like demands, Grobys and Junttila (2021) identify an inverse relation between maximum returns (MAX, henceforth) from the previous week and subsequent CC returns, showing investors' preference for lottery-like CCs. Jia *et al.* (2021) also observe similar behavior using intraday data of the top 84 CCs. However, Lin *et al.* (2021) present contrasting findings, suggesting that a higher MAX leads to a better return in the succeeding month. Since the literature on lottery-like characteristics in CC remains sparse and inconclusive, my study aims to verify previous research findings by analyzing high-frequency intraday data.

The motivation for using high-frequency data stems from another significant development in the same decade as the surge of CCs: the arrival and explosive growth of high-frequency traders. While this technological sophistication increases market efficiency through improved liquidity and price discovery, it also introduces unique intraday trading dynamics. Thus, this question of lottery preferences among CC investors is crucial for

deciphering intraday market efficiency and the influence of the growing number of high-frequency traders.

In contrast to the existing literature in the CC market, which typically relies on daily and monthly data to understand lottery-like preferences among investors, my study delves deeper into high-frequency data (5 min). Therefore, my study design presents a unique prospect to examine [Bali et al.'s \(2011\)](#) (BCW or BCW's, henceforth) "preference for lottery stocks" justification of the MAX effect at the finer intervals.

I consider the top 100 CCs filtered by market capitalization from January 2020 to December 2022. MAX is the highest 5-min return within an hour. As a robustness measure, I also test my hypothesis on MAX (2 and 3) along with MAX, where MAX 2 (3) is the mean of the two (three) highest 5-min returns in an hour. Using portfolio-level analysis and Fama-Macbeth (FMB, henceforth) cross-section regressions, I find CCs with higher MAX give significantly smaller future returns. This finding aligns with existing evidence indicating that lottery CCs are often overvalued and offer lower subsequent returns.

Due to the strong correlation between MAX return, idiosyncratic volatility (IVOL, henceforth) and skewness (SKEW, henceforth), I investigate the robustness of MAX in their presence and unravel that the MAX effect is robust to IVOL and skewness (total, idiosyncratic and coskewness [1]). Additionally, the findings of all measures of skewness are significantly and negatively associated with expected returns on the CC market side, as seen by [Jia et al. \(2021\)](#). Furthermore, IVOL, on its own, shares a negative relationship with expected returns, supporting the puzzling results of [Ang et al. \(2006, 2009\)](#). This contradicts existing studies investigating this relationship in the CC market ([Bouri et al., 2022](#); [Burggraf and Rudolf, 2021](#); [Zhang and Li, 2020](#)), but once controlled for MAX, the relationship reverses or becomes insignificant. Some may question whether the negative association between MAX and expected returns stems from short-term market reversals. Nevertheless, the findings remain robust even after accounting for short-term reversal in the regression model, with the MAX effect retaining its economic and statistical significance.

The study makes the following key contributions. First, it extends the investigation of the anomalous association between MAX and the expected return to the economically significant CC market. Employing portfolio-level and FMB cross-section regression analysis, consistent with earlier studies, I note a negative association between MAX and future return. Previous studies on MAX in CCs primarily relied on daily, weekly or monthly frequency forecast horizons ([Grobys and Junttila, 2021](#); [Jia et al., 2021](#); [Ozdamar et al., 2021](#)), which inherently smooths out critical market microstructures and masks nuanced investor behaviors, creating a significant gap in understanding rapid price dynamics in CC market. The choice of intraday data is justified by the prevalence of high-frequency trading in the CC markets and CC literature, which indicates that CC returns are short-memory processes ([Grobys et al., 2020](#)).

In the context of the CC market, the high-frequency data becomes all the more significant for several reasons. First, CCs are known for their extreme intraday volatility, creating significant opportunities for gains and losses. Previous research on the MAX effect research has noted that high-MAX stocks tend to be highly volatile. Second, volatility clustering extends beyond daily time series and has been observed in intraday returns, strengthening the suggestion that lottery preferences may manifest as a significant intraday phenomenon. Third, algorithmic trading dominates CC markets, with studies indicating that over 70%–80% of CC trading volume is executed through high-frequency trading algorithms (xx), making the intraday data instrumental in untangling investor behavior and fine-tuning investment strategies.

Price anomalies identical to the MAX effect are attributed to investors' inclination toward lottery-like investments ([Annaert et al., 2013](#); [Bali et al., 2011](#); [Cheon and Lee, 2018, 2018](#);

Walkshäusl, 2014). These preferences are linked to behavioral biases such as overreaction and herding, which get smoothened out at daily or lower-frequency price data that largely reflect economic fundamentals and information influxes. The high-frequency data offers several advantages: it offers a more thorough view of market movements, enabling capturing sentiment shifts promptly and allows the development of more accurate predictive models incorporating intraday price patterns and algorithmic trading behaviors.

Though Jia *et al.* (2021) employ intraday data to construct the MAX variable in the CC market, this study diverges and is an improvement from theirs along the following lines. First, they construct daily MAX using the 5-min data, whereas in this study, hourly MAX is calculated, and the analysis is performed on hourly frequency in contrast to daily frequency. Second, their study focuses on delineating how MAX influences the predictability of higher return moments, where they very briefly touch upon the cross-sectional predictability of MAX for the next days' CC returns. They conclude that CC investors portray gambling tendencies and have lottery-like preferences. In contrast, this study provides a more comprehensive analysis at hourly frequency, offering deeper insights into investor behavior and market efficiency in CC markets.

The second contribution of this paper is exploring the association between IVOL and future returns. Contrary to the prevailing literature in CC, the study unravels an IVOL puzzle indicating the negative premium associated with IVOL at high frequency (Bouri *et al.*, 2022; Burggraf and Rudolf, 2021; Zhang and Li, 2020).

The third contribution, perhaps the most exciting, is to untangle the link between MAX, IVOL and skewness in the CC market. This aspect is novel in my study and enriches the existing literature. The study pioneers the examination of MAX robustness to IVOL and various skewness measures at the intraday frequency. BCW in the US market has shown that incorporating MAX as a control variable reverses the negative relationship between IVOL and returns. Similarly, Annaert *et al.* (2013) and Walkshäusl (2014) in the European market indicate that the MAX absorbs the negative IVOL coefficient. Conversely, Nartea *et al.* (2017) and Wan (2018) indicate that IVOL and MAX are independent in the Chinese stock market. In the CC market, Li *et al.* (2021) find a positive risk premium associated with MAX, and this relationship remains robust to IVOL.

Further, considering the substantial involvement of professional capital in the CC market, the study has several practical implications. For instance, money managers could develop trading strategies by taking long (short) positions on low (high) MAX CCs, potentially yielding higher returns. Finally, the study also adds to some of the critical and emerging strands of literature, such as the market efficiency debate in the CC market (Corbet *et al.*, 2019; Tripathi *et al.*, 2022), cross-sectional predictability of the CC returns, and adds a new factor to the CC price determination (Groby and Sapkota, 2019; Liu *et al.*, 2020). Finally, CCs, as such, do not have any intrinsic value, and thus, investor behavior plays a fundamental role in shaping the dynamics of the CC markets. The findings of my study address a very popular debate in the traditional equity market and conclude that the presence of lottery-like preference in CC markets is analogous to equity and other traditional markets.

The remaining paper is structured as follows. Section 2 delves into the related literature and formulates the testable hypotheses. Section 3 describes the data and variable construction; Section 4 reports the primary findings, and the Section 5 summarizes with the key points.

2. Literature review and hypothesis development

2.1 MAX effect

Using extreme returns as a proxy for lottery-like preferences, BCW finds that stocks with the highest daily return in the preceding month experience significantly lower returns in the

subsequent month. This phenomenon, known as the MAX effect, is supported by theoretical explanations provided by [Barberis et al. \(2008\)](#), [Brunnermeier and Parker \(2005\)](#) and [Tversky and Kahneman \(1992\)](#). Similar evidence is found in various international equity markets, e.g. [Annaert et al. \(2013\)](#), [Barinov \(2018\)](#) and [Walkshäusl \(2014\)](#) investigated the BCW's MAX hypothesis in several European countries and supported their findings [2]. Several studies have attempted to modify and enhance the MAX measure introduced by BCW to make it more robust. Examining the MAX and expected weekly returns relationship, [Cao and Han \(2016\)](#) define MAX (5) as the mean of the top five daily returns within a rolling 30-day timeframe.

Moreover, considering the daily price limits in the equity market, [Hung and Yang \(2018\)](#) introduce a “modified-MAX measure” and concur with previous research findings, suggesting that stocks with the highest modified MAX are often overvalued and give lower returns than stocks with the lowest modified MAX. Expanding on this, [Wang and Lien \(2023\)](#) investigate the modified-MAX measure's impact on the Taiwan stock market, considering factors like investors' limited attention and the 52-week high price as an anchoring point. [Fong and Toh \(2014\)](#) contend that the MAX effect correlates with investor sentiment, leading to the underperformance of stocks with high MAX values, in contrast to the outperformance of stocks with low MAX values. Building on this, [Byun et al. \(2023\)](#) note that lottery preferences are more dominant during the economic slowdown when stocks with high MAX returns are drawing more attention. [Zhong and Gray \(2016\)](#) in the Australian and [Goh and Kim \(2024\)](#) in the South Korean market illustrate that the MAX effect is notable in the most overvalued stocks but is reversed in the case of the most undervalued stocks. Moreover, regarding behavioral mechanisms, [Baars and Mohrschladt \(2024\)](#) propose that the short-range return expectedness of MAX is determined by investor overreaction [in line with [Kahneman and Tversky, 1979](#); [Odean et al., 1998](#))] rather than CPT preferences as claimed by the previous studies. Investigate the role of weather-induced mood in lottery preferences [Bradrania and Gao \(2024\)](#) find that the MAX effect mainly exists following sunny weather, which indicates pleasant weather conditions.

Contrary to its frequent negative effect, several studies report that stocks with MAX returns also yield significant returns in subsequent periods ([Aboulamer and Kryzanowski, 2016](#); [George and Hwang, 2004](#); [Li et al., 2021](#); [Lin, 2023](#)). The intuition behind this is that for stocks that have given the top returns, investors are less likely to see them generating very high payoffs in the immediate future. In contrast, stocks that have performed poorly are more likely to be viewed by investors as akin to lotteries, as their poor performance suggests the potential for a strong rebound and the opportunity to generate substantial gains ([Birru and Wang, 2016](#)).

Given the distinct characteristics of CCs, characterized by their heavy-tailed and extreme return behavior ([Baur et al., 2018](#); [Cheah and Fry, 2015](#); [Enoksen et al., 2020](#)), the presence of lottery-like preferences among investors in this market becomes an intriguing research topic. Despite the well-established significance of higher moments in asset pricing models and the extensive research on idiosyncratic risk, there has been limited investigation of these factors within the context of the CC market. In particular, very few studies have tested the existence of the “MAX effect” among CC investors. Existing research on this subject has produced conflicting findings. [Groby and Junttila \(2021\)](#) modified the MAX measure developed by BCW to a weekly forecast horizon and found evidence of equity-like behavior within the CC market. They report that the mean gap in raw and risk-adjusted returns between CCs in the lowest and highest MAX quantiles is positive. Similar findings are reported by [Jia et al. \(2021\)](#) and [Kumar Kulbhaskar and Subramaniam \(2022\)](#).

However, contrasting results have been observed by [Li et al. \(2021\)](#), noting the opposite behavior and naming it “MAX momentum.” Similar findings were reported in studies by [Lin et al. \(2021\)](#) and [Ozdamar et al. \(2021\)](#). Their findings challenge the traditional inverse MAX-return relationship and underscore the complexity of investor sentiments and CC market dynamics.

Thus, the literature on the MAX effect in the CC market is scarce and inconclusive. In this study, the MAX effect is examined using the high-frequency intraday data, and the choice is justified because of the prevalence of high-frequency trading in the CC markets and CC returns being the short-memory processes ([Grobys et al., 2020](#)). Thus, the following hypothesis is proposed:

H1a. There is a significant relationship between CC returns and MAX.

The positive relationship between the MAX effect and returns at the intraday level can be attributed to the high volatility and speculative nature of CC markets, which amplifies investors’ risk-taking behavior. Conversely, high-frequency trading and the unique dynamics of CC markets might lead to an inverse relationship between the MAX effect and returns, as frequent price adjustments and market efficiency at shorter intervals may reduce the potential for high returns associated with lottery-like investments.

2.2 IVOL and MAX

Following [Ang et al.’s \(2006\)](#) seminal work, several studies have researched the link between IVOL and subsequent stock returns ([Bali et al., 2011, 2017, 2019](#); [Cao et al., 2022](#); [Walkshäusl, 2014](#)). However, the empirical findings remain inconclusive. [Merton \(1987\)](#) posited that investors demand higher returns to reward for the added risk of having undiversified portfolios. In line with this, studies like [Barberis and Huang \(2001\)](#), [Goyal and Santa-Clara \(2003\)](#) and [Fu \(2009\)](#) report a positive link between IVOL and following returns. Conversely, studies such as [Ang et al. \(2006\)](#) and [\(2009\)](#) observe a negative link between IVOL and stocks’ returns in a similar sample period. While various classical paradigms can explain the positive relationship, the negative association remains more perplexing and, hence, the IVOL puzzle. [Stambaugh et al. \(2015\)](#) elucidated the negative IVOL-return relationship by introducing arbitrage asymmetry and risk concepts. They argued that the negative IVOL effect is particularly prominent in overpriced stocks due to arbitrage asymmetry, which outweighs the positive IVOL effect in underpriced stocks, resulting in the observed puzzle. [Wang et al. \(2023\)](#) propose disagreement and speculative trading as an essential driving force behind the IVOL puzzle. Similar investigations have been carried out across various asset classes, including the bond market ([Chung et al., 2019](#)), forex ([Guo and Savickas, 2008](#)) and commodity futures ([Fernandez-Perez et al., 2018](#)) and ETFs ([Duanmu et al., 2024](#)).

More recently, the debate has also extended to CC markets, though the evidence remains limited. Investigating the pricing of IVOL in the cross-section of CC returns, [Zhang and Li \(2020\)](#) demonstrate a positive association involving IVOL and expected CC returns, which is robust to accounting for factors like momentum, liquidity, skewness, etc. Similarly, [Bouri et al. \(2022\)](#) conclude that CC markets are more efficient than traditional financial markets, and IVOL is appropriately priced, with investors receiving a premium for taking the additional risk. Using long-short portfolios for 1000 CCs from 2013 to 2019, [Burggraf and Rudolf \(2021\)](#) suggest that high-volatility CCs yield a higher return than low-volatility ones. Similar findings are reported by [Li et al. \(2021\)](#). The results show that the CC market does not exhibit the same anomalies as conventional financial markets. In my study, I contribute to the literature on CC return volatility and test for the existence of the IVOL effect at intraday frequency.

Additionally, since IVOL is considered one of the features of a lottery-like asset class (Kumar, 2009), it can be argued that the negative “MAX effect” I report may be related to IVOL, with MAX acting as a mere proxy. Chichernea *et al.* (2019) also uncovered that the IVOL puzzle is more pronounced in stocks owned mainly by investors with a penchant for lottery-like investments, especially during an economic slowdown when demand for lottery-like payouts increases. Their findings suggest that lottery inclinations could significantly contribute to the IVOL puzzle. Additionally, accounting for MAX alters the negative association between IVOL and returns in the US stock market (Bali *et al.*, 2011), consistent with similar findings in the European market (Walkshäusl, 2014). Conversely, Nartea *et al.* (2017) and Wan (2018) suggest that the MAX and IVOL effects are independent in the Chinese stock markets, with MAX not influencing the IVOL effect. As a result, the proposed hypotheses are:

H2a. IVOL is a significant predictor of CC returns, and IVOL and MAX are independent in the CC market.

2.3 Skewness and MAX

There is overwhelming evidence from the expected returns models incorporating skewness that investors consider higher moments in returns when making investment decisions (Kraus and Litzenberger, 1983, 1976; Rubinstein, 1973). In the context of diversification, early models primarily focused on examining how securities’ co-movements with the overall market portfolio affected their pricing. They tested if a security’s coskewness (COSKEW, henceforth) or co-kurtosis with the market influenced expected returns, largely disregarding their idiosyncratic component. For instance, Harvey and Siddique (2000) contend that COSKEW significantly contributes to explaining variations in expected returns among assets, even after accounting for firm factors like market capitalization and book-to-market ratio. Dittmar *et al.* (2002) also investigated the influence of a security’s COSKEW and co-kurtosis with the market portfolio on investors’ expected returns. Smith (2007) emphasizes the significance of COSKEW in determining equity returns and highlights its time-varying nature. Similar findings are observed in other markets as well, for instance, in the US equity and bond market by Yang *et al.* (2010), in mutual funds by Moreno and Rodríguez (2009) and in the currencies market by Chan *et al.* (2018). Dong *et al.* (2022) demonstrate a significant negative association between COSKEW and market return by investigating COSKEW pricing in the local, regional and world markets. However, these models assume that diversification erodes skewness exposure, thus focusing only on the COSKEW.

However, recent models and empirical studies have placed greater emphasis on the higher moment properties of individual securities. Specifically, they suggest that idiosyncratic skewness (ISKEW, hereafter) in a security’s return distribution may also play a significant role in understanding returns (Barberis and Huang, 2008; Conrad *et al.*, 2013; Mitton and Vorkink, 2007). These models assume that investors maintain an under-diversified portfolio to capitalize on return skewness, indicating a preference for lottery-like investments (Boyer *et al.*, 2010; Brunnermeier *et al.*, 2007). My motivation stems from theoretical arguments presented in the literature. For instance, Barberis and Huang (2008) posit that investors have CPT preferences and tend to overweigh the extreme outcomes with small probabilities. Similarly, Brunnermeier *et al.* (2007) and Brunnermeier and Parker (2005) argue that investors hold biased assessments of probabilities and tend to be optimistic about future payoffs. In this backdrop, the proposed hypothesis is:

H3a. Skewness is a significant predictor of CC returns, and skewness and MAX are independent in the CC market.

3. Data and method

3.1 Data

The trading data for all CCs is collected from Binance, a prominent global CC exchange [3]. The final sample comprises the top 100 CCs based on market capitalization for 2020, 2021 and 2022 [4]. I have included only those coins with information on available price, volume and market capitalization. The combined market capitalization of these top 100 coins in the sample totals approximately \$2.5tn, representing over 95% of the total CC market. This ensures that the analysis accurately captures the dynamics of the overall market. The data is at a minute-by-minute frequency within the 24/7 CC spot market. To ensure robustness and mitigate potential outliers, I construct returns at a 5-min frequency. It is worth noting that on Binance, the trading volume and price data are presented in terms of CC pairs. Each CC is paired with different quote assets. I have selected USDT pairing for my analysis, meaning that all prices are denominated in Tether (USDT) [5]. For CC i during hour t , the MAX is the interval on which the CC earns its highest 5-min return for the hour. I take equal-weighted mean returns of all the CCs in the sample as the market index [6]. The sample entails approximately 2.1 million CC-hours (i, h) and encompasses MAX events from January 1, 2020, to December 18, 2022. 2020-01-01:00 is the first MAX event hour, and 2022-11-18:01 is the last post-MAX event hour. The variables used in the study are explained in detail in Appendix 1 [7].

3.2 Methodology

The empirical analysis uses portfolio-level and FMB cross-sectional regression analyses to study the relationship between MAX, IVOL and skewness over the past hour and future returns. I conduct portfolio-level analysis using a portfolio-sort test as follows. From January 1st, 2020, to November 18th, 2022, at the beginning of each hour, I sort the cross-sectional observations (CCs) into quantiles based on a specific variable of interest (MAX, IVOL or SKEW, etc.). Portfolio 1 consists of CCs with the lowest values of the chosen variable over the past hour, while Portfolio 5 includes CCs with the highest values. Next, I calculate the mean return for each portfolio quantile in the subsequent hour, generating a time series of returns for each quantile. Ultimately, I use these time series to estimate the average return for each quantile across the full sample period.

Additionally, I examine the relationship at the individual CC level using the FMB methodology. This approach enables to simultaneously consider multiple control variables, which is a distinct advantage compared to portfolio sorts. Hourly cross-sectional regressions are performed, where I regress the returns of individual CCs for that hour against lagged explanatory variables. I then provide time-averaged coefficients for these independent variables. FMB offers several advantages. First, it is well-suited for unbalanced panels, making it a robust choice for data sets like mine. Second, it provides flexibility by allowing me to incorporate time-varying betas. Finally, by conducting cross-sectional regressions for each time interval, the intercept absorbs and estimates the time effect, resulting in significantly smaller residual variability compared to OLS and thus greater efficiency.

4. Empirical results and analyses

4.1 Summary statistics

Table 1 reports summary statistics and correlation coefficients for the key variables in the analysis. MAX has a mean of 0.62% and a median of 0.47%. IVOL has a mean of 0.23% (median of 0.17%), and the average skewness (SKW) is -0.0073. MAX and IVOL also demonstrate a high skewness and kurtosis. Variants of MAX [MAX (1), MAX (2), MAX (3)] show a very high correlation among themselves. The correlation between MAX and IVOL is

Table 1. Summary statistics: the table presents average cross-sectional summary statistics of the main variables employed in the study

	MAX (1)	MAX (2)	MAX (3)	Beta	Reversal	Mom	Illiq	Log vol	IVOL	SKW
<i>Panel A: Summary statistics</i>										
Min.	0.0001	0.0001	0.0001	-2.6735	-0.0125	-0.0186	0.00	0.9813	0.0001	-1.8666
1st Qu.	0.0029	0.0025	0.0021	0.519	-0.0016	-0.0026	0.0224	8.047	0.0011	-0.4357
Median	0.0047	0.0039	0.0033	0.9665	0.00	0.00	0.0558	10.0332	0.0017	0.00
Mean	0.0062	0.0051	0.0043	0.9962	0.00	0.00	0.0918	10.0393	0.0023	-0.0073
3rd Qu.	0.0077	0.0063	0.0053	1.4124	0.0016	0.0025	0.1162	12.2244	0.0028	0.4218
Max.	0.0303	0.0236	0.0197	5.1009	0.0128	0.0191	0.6861	19.0558	0.0114	1.8549
SD	0.0052	0.0041	0.0034	1.069	0.0037	0.0056	0.1129	3.3533	0.0019	0.7028
SK	2.1917	2.0878	2.054	0.3718	0.0316	0.0238	2.778	-0.1033	2.2739	-0.0038
Kurt	9.0478	8.5085	8.3493	6.7219	5.6813	5.4068	12.6923	3.2787	9.394	3.2786
<i>Panel B: Correlation matrix</i>										
MAX (1)	1									
MAX (2)	0.978***	1								
MAX (3)	0.955***	0.992***	1							
Beta	0.176***	0.181***	0.182***	1						
Reversal	0.077***	0.091***	0.105***	-0.011***	1					
Mom	0.118***	0.135***	0.150***	-0.014***	-0.003***	1				
Illiq	0.385***	0.389***	0.385***	0.054***	-0.001	0.001	1			
Log vol	0.185***	0.192***	0.193***	0.048***	0.002**	0.001	-0.229***	1		
IVOL	0.786***	0.797***	0.791***	0.122***	0.006***	0.016***	0.392***	0.191***	1	
SKW	0.331***	0.271***	0.228***	0.001	0.040***	0.088***	0.007***	0.001	0.067***	1

Note(s): These statistics include sample mean, standard deviation (SD), median, skewness (SK), kurtosis (Kurt), minimum (min.), maximum (max.) and correlation coefficients. All the variables are winsorized at the 0.5 and 99.5 percentiles. $MAX(n)_{ij}$ is calculated as the average of n maximum 5-min returns for CC i during the hour $t-1$ (where $n = 2, 3$), the lag and lead of the market portfolio and the current market are used when estimating beta (Dimson, 1979; Scholes and Williams, 1977). illiquidity, following Amihud (2002), is equal to the ratio of the absolute return over traded volume (in USD) (scaled up by 10^3). IVOL is the standard deviation of the residuals from the Fama-French market model. Momentum (mom) is the cumulative return for CC i for 15 min from $t-3$ to $t-2$ (Jegadeesh and Titman, 1993; Liu and Tsyvinski, 2021). The return of the most recent 5 min is defined as a short-term reversal (Jegadeesh and Titman, 1993). Skewness is computed as total Skewness (the natural measure of the third central moment) using returns over the previous hour

Source(s): Authors' own work

notably high at 78.6%, and between MAX and SKW is 33.1%. This strong correlation aligns with prior research (Annaert *et al.*, 2013; Bali *et al.*, 2011; Walkshäusl, 2014). The high degree of correlation between the variables could potentially cause multicollinearity issues in regression analyses, leading to unstable regression coefficients. However, this concern is alleviated in my study, as indicated by the Variance Inflation Factor (VIF) for MAX consistently remaining below 2 across all scenarios [8].

4.2 CC Attractiveness based on MAX

4.2.1 Portfolio analysis. The primary objective is to examine whether MAX adds to the determination of the cross-section of CC returns. To accomplish this, I employ standard asset pricing tests to explore the link between average CC returns and MAX and see if the findings are uniform with the theoretical framework of BCW. Here, I initially examine how mean returns differ across CCs with varying degrees of MAX. Subsequently, I employ the FMB methodology to analyze the impact of MAX in cross-sectional regressions.

I initiate my analysis with a portfolio-level examination akin to BCW, revealing a pronounced MAX effect in the CC market. At the conclusion of each hour (t), I rank CCs based on their MAX returns (N), where $n = 1, 2$ and 3 , in ascending order. Portfolio 5 comprises CCs with the highest MAX returns, while Portfolio 1 includes those with the smallest MAX return. I create a zero-cost top-minus-bottom portfolio by taking long positions on CCs in Portfolio 1 and short positions on CCs in Portfolio 5. Using the equal-weighted average returns of the hour ($t + 1$), I offer portfolio-level proof for the MAX effect in the CC market.

I estimate portfolio returns using the average returns and Capital Asset Pricing Model (CAPM, henceforth) alpha [9] over the sample from January 2020 to December 2022. Table 2 illustrates the portfolio-level demonstration of the MAX effect in the CC market. Portfolio returns gradually decline from Portfolio 1 to Portfolio 4, but a sharp decrease is evident in Portfolio 5, dropping from 0.0034% to -0.0066%. Thus, implementing the zero-cost trading strategy based on the MAX (1) effect offers a statistically significant return differential of 0.0610% at the 1% level, with a t -value of 19.46. This indicates a robust MAX effect in the CC market, suggesting that CCs with high MAX returns tend to underperform compared to those with lower MAX returns. These findings suggest that CCs with the highest MAX, exhibiting a lottery-like property, may appeal to investors but experience negative performance in subsequent periods. Such trading behavior underscores recent findings on CC research, highlighting the speculative nature of CC markets over their utility as a medium of exchange.

The return and alpha results in Table 2 fundamentally align with what Grobys and Junttila (2021) and Jia *et al.* (2021) document in the CC market. However, it is crucial to note that my study covers a different sample period (Jan. 2020 – Dec. 2022) and uses intraday data instead of the weekly or daily data employed in previous studies. This distinction in both sample period and data frequency may account for variations in return and CAPM alpha results.

While employing a single 5-min period with the MAX return as a proxy for lottery-like returns seems intuitive, it can be somewhat subjective. As a robustness test, I also rank CCs based on the mean of the N (where $n = 1, 2$ and 3) highest 5-min returns within the hour, and the results are presented in Table 2. Additionally, I report the difference between the average returns and CAPM alphas between Portfolios 1 and 5. As I increase the averaging to $n = 2$ and 3 , the raw return and CAPM alpha differences exhibit greater economic and statistical significance, thus establishing the robustness of the findings.

4.2.2 Determinants of MAX. Next, the question is, what is it about a high MAX portfolio that makes it attractive to investors? To understand this, I undertake two approaches. First, I

Table 2. Univariate MAX sorts: Each hour, I sort CCs based on MAX(N), five portfolios are formed every hour from 2020-01 to 01:00-2022-11-18:00 by sorting CCs based on the MAX (N) over the past hour. Portfolio 1 (5) is the portfolio of CCs with the lowest (highest) MAX over the past 1 h

	Low	MAX2	MAX3	MAX4	High	Low-High
<i>Panel A: MAX (1)</i>						
Excess return	0.0265*** (5.74)	0.0227*** (3.76)	0.0178*** (2.81)	0.0151** (2.28)	-0.0345*** (-5.00)	0.0610*** (19.46)
CAPM alpha	0.026*** (5.79)	0.023*** (3.84)	0.018*** (2.80)	0.015** (2.31)	-0.034 *** (-5.50)	0.0600*** (17.12)
<i>Panel B: MAX (2)</i>						
Excess return	0.0274*** (5.96)	0.0239*** (3.98)	0.0201*** (3.17)	0.0142** (2.14)	-0.0368*** (-5.30)	0.0641 *** (19.85)
CAPM alpha	0.027*** (5.94)	0.024*** (3.95)	0.02*** (3.25)	0.014** (2.15)	-0.036*** (-5.85)	0.0640*** (17.71)
<i>Panel C: MAX (3)</i>						
Excess return	0.0291*** (6.35)	0.0248*** (4.14)	0.0207*** (3.27)	0.0129* (1.93)	-0.0386*** (-5.54)	0.0676*** (20.70)
CAPM alpha	0.0290*** (6.47)	0.0250*** (4.14)	0.0210*** (3.42)	0.0130* (1.96)	-0.0380*** (-6.08)	0.0670*** (18.80)

Note(s): The table presents the average hourly returns and CAPM alphas for the hedge portfolio, which involves going long on the high MAX portfolio and short on the low MAX portfolio. The last column displays the differences in returns and alphas relative to the CAPM model between portfolios 1 and 5, alongside the corresponding Newey West adjusted t-statistics. Average raw and risk-adjusted returns are expressed in percentage terms. HAC-robust t-statistics based on Newey and west (1987) standard errors are provided in parentheses. Statistical significance is indicated by *, ** and ***, representing significance levels at 10, 5 and 1%, respectively

Source(s): Authors' own work

determine the cross-sectional median averages of portfolio characteristics sorted based on MAX (1). Second, I run a cross-sectional regression of MAX on CC characteristics. Panel A, Table 3 presents the summary statistics for the CCs across different portfolios. This includes the maximum return (in percent), illiquidity measure (scaled by 10^3), reversal, the cumulative return for CC i for 15 min from $t-3$ to $t-2$ (momentum), the IVOL, beta, total skewness and log of the trading volume. Here, I calculate the median of the cross-section of CC attributes for each portfolio every hour and report the time-series mean of these medians. I observe that high MAX CCs demonstrate greater market betas, along with higher momentum, higher reversal and higher log of the trading volume. This pattern of relationships largely mirrors the findings in equity markets where BCW documents similar patterns in US stocks; Walkshäusl (2014) in the European markets and Nartea *et al.* (2017) in the Chinese stock market, where high MAX stocks show higher betas, higher skewness, more illiquidity and greater positive IVOL. The persistence of MAX, indicated by the coefficient on lagged MAX (0.48), is substantially higher than the approximately 0.40 documented by BCW, reflecting the unique characteristics of CC markets. Since high MAX CCs exhibit other lottery traits such as higher IVOL, more SKEW and greater illiquidity, these characteristics might bias the results toward lower returns. However, in my study, I address this concern by controlling for these variables in the FMB cross-sectional regression, enabling me to ascertain the net effect of MAX on future returns.

Panel B Table 3 displays average coefficient estimates from regressing MAX on various CC characteristics, including beta, momentum, reversal, illiquidity, log volume, IVOL and SKEW. All estimated coefficients exhibit statistical significance. Particularly noteworthy is the positive association between MAX and lagged MAX, beta, IVOL, illiquidity and log trading volume. The positive associations between MAX and key variables like beta (0.0322), IVOL (0.8655), illiquidity (3.5681) and log volume (0.0001) are particularly noteworthy. These relationships are consistent in direction, though more statistically significant than the traditional markets (Bali *et al.*, 2011; Cheon and Lee, 2018). Additionally, the significant coefficient on lag MAX indicates the persistence of MAX over a short horizon. These findings suggest that the composition of the high MAX portfolios in the CC market is consistent with BCW's decile MAX portfolio in the US equity market.

4.2.3 Fama-Macbeth cross-sectional regression. I conduct FMB cross-sectional regressions in this section to examine the MAX effect. The regression model used is:

$$r_{it+1} = \lambda_{0t} + \lambda_{1t}MAX_{it} + \epsilon_{it+1} \quad (1)$$

$$r_{it+1} = \lambda_{0t} + \lambda_{1t}MAX_{it} + \sum \lambda_{it}Control_{it} + \epsilon_{it+1} \quad (2)$$

In the regression model, MAX represents the maximum return in hour t , and r is the CC return of the hour $t+1$. I incorporate several well-recognized return determinants to assess the net MAX effect, including the market beta, reversal and momentum, illiquidity ratio, IVOL, SKEW and log volume. I use Newey–West adjusted standard errors to obtain the associated t -values. The results presented in Table 4 indicate a significantly negative association between MAX and expected returns, both in univariate regression and regression models with multiple controls. It reports the time series averages of the slope coefficients λ_{it} ($i = 1, 2, 3, \dots, 6$). Specifically, in the univariate regression, MAX exhibits a significant negative coefficient of -0.0928 with a t -values of -27.81 . This coefficient reveals an interesting pattern when compared to findings across different markets. In traditional equity markets, BCW documented a relatively modest MAX coefficient of approximately -0.043 in the US, while varying effects have been observed in European markets, with Walkshäusl (2014) reporting

Table 3. Determinants of MAX: Panel A of the table reports the averages of the cross-sectional median of characteristics of portfolios sorted on MAX (1)

Panel A: Summary statistics for portfolios of CCs sorted by MAX

	MAX	Beta	Mom	Reversal	SKEW	Illiq	IVOL	Log vol
Low	0.280	0.598	-0.030	-0.011	-0.300	0.005	0.115	9.620
2	0.414	0.905	-0.017	-0.001	-0.156	0.006	0.151	9.730
3	0.523	1.031	-0.006	0.004	-0.024	0.007	0.181	9.944
4	0.674	1.155	0.005	0.008	0.113	0.007	0.233	10.176
High	1.056	1.386	0.034	0.012	0.369	0.010	0.397	10.808

Panel B: Cross-sectional regression of MAX on CC characteristics

	Mod1	Mod2	Mod3	Mod4	Mod5	Mod6	Mod7	Mod8
Lag max	0.4814*** (133.49)	0.4612*** (116.91)	0.4607*** (117.26)	0.4577*** (114.22)	0.43*** (123.23)	0.4102*** (143.67)	0.4823*** (178.30)	0.1286*** (32.72)
Beta		0.0352*** (16.30)	0.0315*** (16.90)	0.0317*** (17.03)	0.0304*** (17.09)	0.0282*** (17.02)	0.01874*** (14.13)	0.0322*** (22.63)
Mom			-0.0211*** (-14.225)	-0.0253*** (-15.536)	-0.0202*** (-13.364)	-0.0174*** (-11.876)	-0.0124*** (-8.68)	0.0162*** (10.65)
Reversal				-0.0137*** (-6.47)	-0.0072*** (-3.66)	-0.0048*** (-2.59)	-0.0039*** (-2.15)	0.0261*** (13.39)
Illiq					4.4794*** (35.30)	5.9148*** (40.68)	4.9738*** (38.61)	3.5681*** (32.16)
Log vol						0.0001*** (20.66)	0.0001*** (13.78)	0.0001*** (10.84)
SKW							-0.0896*** (-52.78)	-0.0308*** (-23.92)
IVOL								0.8655*** (78.71)
Adj. R Sq.	23.329	25.225	26.816	28.275	29.817	30.377	32.228	32.228

Note(s): For each hour t , the median of the cross-section of CC characteristics for each portfolio is computed. The time-series averages of these medians are reported. Beta is the market beta from the three-factor model estimated maximum return (MAX) in the past hour, a measure of illiquidity (scaled by 10^3), legged return (reversal), the cumulative return for CC i for 15 min from $t-3$ to $t-2$ (mom), the idiosyncratic volatility (IVOL), market risk (beta), total skewness (SKEW) and log of trading volume. Panel B reports the cross-sectional predictability of MAX. From 2020-01 to 01:00-2022-11-18:01, I conduct a CC-level cross-sectional regression every hour. Beta and skewness are scaled by 10^2 . This regression analyses the maximum daily return within that hour (MAX) and lagged predictor variables, including the MAX from the previous hour and other control variables, as defined in section 3. The table reports the time-series averages of the cross-sectional regression coefficients, their associated Newey and West (1987) adjusted t -statistics (in parentheses) and the regression R -squares (%)

Source(s): Authors' own work

Table 4. Cross-sectional regressions MAX and controls: for each hour from 2020–01 to 01:00–2022–11–18:01, I conduct hourly cross-sectional regression of the return in that hour on subsets of lagged predictor variables, including MAX in the previous hour and six control variables

	Mod1	Mod2	Mod3	Mod4	Mod5	Mod6
MAX	−0.0928*** (−27.81)	−0.1049*** (−31.29)	−0.0985*** (−28.91)	−0.0884*** (−25.96)	−0.0833*** (−23.36)	−0.0830*** (−23.16)
Beta		0.0249*** (14.01)	0.0204*** (11.06)	0.0197*** (10.00)	0.0183*** (9.30)	0.0173*** (8.82)
Mom			−0.0592*** (−18.358)	−0.0802*** (−22.88)	−0.0742*** (−20.99)	−0.0718*** (−20.23)
Reversal				−0.1699*** (−31.948)	−0.1628*** (−31.528)	−0.1602*** (−31.02)
Illiq					−0.0698 (−0.52)	−0.0654 (−0.45)
Log vol						0.0001 (1.47)
Adj. R Sq.	4.733	7.496	9.74	12.056	13.093	13.318

Note(s): The set of control variables includes market beta (BETA), momentum (MOM), illiquidity (Illiq), short-term reversal (reversal) and log volume. Beta and log volume are scaled by 10². This procedure yields a time series of slope coefficients. In each row, the table presents the time-series averages of the cross-sectional regression slope coefficients and their associated Newey and West (1987) adjusted t-statistics (in parentheses)

Source(s): Authors' own work

coefficients of -0.047 and -0.069 for equal-weighted and value-weighted cross-sectional regressions in the integrated European market. [Annaert et al. \(2013\)](#) found a substantially smaller effect with a coefficient of -0.0004 across 13 European countries. In Asian markets, the MAX effect shows considerable variation, with [Hung and Yang \(2018\)](#) documenting a coefficient of -0.078 in the Taiwan stock market, while [Nartea et al. \(2017\)](#) found a much stronger effect in the Chinese stock market with a coefficient of -0.155 . In the CC market context, [Groby and Junttila \(2021\)](#) documented an even stronger effect with a coefficient of -0.1938 . In Mod6 ([Table 4](#)), where I control for a set of variables, for one standard deviation increase in its MAX, a CC's return decreases by 0.043% relative to other CCs [[10](#)].

Additionally, the MAX coefficient remains highly significant even after including beta and momentum variables, indicating minimal variation in its size. While CCs with large positive hourly returns may also exhibit the reversal phenomenon [[11](#)], I mitigate this concern by accounting for short-term reversal in my study. The addition of the reversal component in the model marginally diminishes the MAX coefficient; nevertheless, it retains its statistical and economic significance. Moreover, even after incorporating all return determinants, the result remains significant (-0.0830 , t -values = -23.16). In summary, the results obtained from the FMB cross-sectional regressions are consistent with the portfolio-level evidence presented in Section 4.2.1. This further supports my hypothesis (H1a) regarding the significant MAX effect in the CC market.

4.2.4 Persistence of MAX. Investors would pay a premium for CCs with high MAX returns only on the anticipation that this performance will also be repeated in the future. Further, past MAX is not necessarily a good proxy for expected future MAX and does not reliably predict future MAX. I give two evidence in support of this claim. First, I construct an hour-to-hour portfolio transition matrix, showing the average likelihood that a CC in the portfolio i in one hour will be in the portfolio j in the next hour. If MAX is completely chance and non-persistent, all odds are approximately 10%, indicating that the current MAX offers no insights into future MAX. [Table 5](#) shows that the diagonal probabilities are above 20% in all the portfolios and are most robust in extreme portfolios (1 and 5). For instance, CCs in Portfolios 1 and 5 have approximately a 45% chance of remaining in the same portfolio in the next hour. However, it remains above 20% comfortably in all the other instances. Moreover, the persistence property of MAX remains intact even if I construct the transition matrix over a longer horizon ($t+3$, $t+6$ and $t+12$). These findings align with [Zhong and Gray \(2016\)](#) in the Australian stock market.

Furthermore, the cross-sectional regressions in Panel B [Table 3](#) provide additional evidence of the persistence of MAX returns. These regressions examine MAX in relation to lagged predictor variables, including lagged MAX. For each hour in the sample, a regression is conducted across CCs based on MAX within that hour, incorporating the MAX from the previous hour and other lagged control variables. In the univariate regression (MAX on lagged MAX), the MAX coefficient is positive with a magnitude of 0.4814 and a t -values of 133.49 is notably larger than coefficients reported in equity market studies. For instance, BCW and [Annaert et al. \(2013\)](#) report the coefficient of 0.4054 and 0.36 for US and European markets, respectively, for monthly MAX persistence in stocks. 0.207 (The MAX effect: European evidence). The findings are also more robust than those reported in the Taiwanese stock market.

Furthermore, the R -squared value exceeding 23% shows a considerable level of explanatory power across the cross-section, implying that CCs demonstrating exceptionally positive returns in 1 h also maintain similar characteristics in the subsequent hour. Even after integrating other control variables into the model, the coefficient on lagged MAX retains its significance, underscoring the persistence of MAX returns.

Table 5. Persistence of MAX: At each hour t , I categorize CCs into 5 portfolios based on MAX, ensuring an equal number of CCs in each portfolio. In hour $t + 1$, I repeat this process. Then I check to which portfolio CCs have moved. The table shows average transition probabilities, indicating the likelihood of a CC moving from portfolio i (rows) to portfolio j (columns) in the hours $t + 1$, $t + 3$, $t + 6$ and $t + 12$. MAX (1) is the highest 5-min return over the past hour t

	Low (%)	2 (%)	3 (%)	4 (%)	High (%)
<i>Panel A: $t + 1$</i>					
Low	44.5	22.5	16.3	11.8	7.4
2	21.4	25.7	22.5	17.7	10.5
3	15.5	22.8	24	22	15
4	11	17.6	21.7	25.1	22.5
High	7.5	11.3	15.6	23.3	44.5
<i>Panel B: $t + 3$</i>					
Low	43.5	22.3	16.4	12.3	8.2
2	20.9	25.1	22.1	17.8	11.8
3	15.6	22.3	23.4	21.8	16.1
4	11.5	17.8	21.4	24.3	23
High	8.5	12.5	16.7	23.8	40.9
<i>Panel C: $t + 6$</i>					
Low	42.8	22.1	16.5	12.6	8.8
2	20.7	24.8	22	18	12.3
3	15.7	22	23.1	21.8	16.7
4	11.8	17.9	21.3	23.9	23
High	9.1	13.2	17.2	23.7	39.1
<i>Panel D: $t + 12$</i>					
Low	42.3	21.9	16.6	12.7	9.2
2	20.5	24.3	21.8	18.3	12.9
3	15.7	22	22.8	21.7	17.1
4	12	17.9	21.2	23.5	23.2
High	9.5	13.9	17.6	23.8	37.6

Source(s): Authors' own work

The findings highlight that while MAX persistence exists in both traditional and CC markets, the effect is considerably stronger in CCs, particularly at high-frequency horizons. This suggests that the gambling preferences and attention-grabbing hypothesis documented in stock markets by [Kumar \(2009\)](#) and [Barber and Odean \(2008\)](#) may be amplified in CC markets.

4.3 MAX and the IVOL puzzle

This section delves into the relationship between MAX and IVOL in the CC market. Conventional wisdom would imply that since idiosyncratic risk can be diversified easily, it either does not warrant a risk premium or commands a positive risk premium ([Merton, 1987](#)). However, [Ang et al. \(2006, 2009\)](#) state that IVOL exhibits negative pricing in the cross-section, representing that stocks with high IVOL tend to yield lower subsequent returns. I explore how IVOL is priced in the CC market through cross-sectional regressions of CC returns on IVOL over the previous hour. The following equations are estimated:

$$r_{it+1} = \lambda_{0t} + \lambda_{1t}IVOL_{it} + \epsilon_{it+1} \tag{3}$$

$$r_{it+1} = \lambda_{0t} + \lambda_{1t}IVOL_{it} + \sum \lambda_{it}Control_{it} + \epsilon_{it+1} \quad (4)$$

In the regression model, IVOL represents the idiosyncratic volatility in hour t , and r is the CC return of the hour $t+1$. Table 6 displays the average coefficient estimate from these regressions. Across all regression specifications (Mod1 to Mod7), I consistently observe significantly negative coefficient estimates on IVOL, ranging from -0.1543 to -0.1722 on average, all with t -statistics exceeding -19 . This negative association suggests that CCs characterized by high IVOL tend to yield notably lower future returns, thus affirming the presence of a robust IVOL puzzle within CC markets. The findings align with the evidence of the IVOL puzzle in various equity markets (Ang *et al.*, 2006, 2009; Annaert *et al.*, 2013; Bali *et al.*, 2011; Walkshäusl, 2014). Ang *et al.* (2009) report coefficients around -2.014 for US stocks and -1.536 when pooled for all developed countries, while Walkshäusl (2014) finds that the average IVOL coefficient estimates range from -0.687 to -0.853 in equal-weighted returns and from -1.243 to -1.376 in value-weighted returns. This suggests a stronger IVOL puzzle in CC markets at intraday frequency.

To address concerns regarding MAX possibly serving as merely a proxy for IVOL without adding new information, I explore the relationship between MAX and IVOL using two distinct approaches. First, I explore the effect of adding MAX to IVOL in the cross-section of mean returns. The regression model (8) in Table 6 shows that once controlled for the MAX in the model, the IVOL puzzle gets reversed, and IVOL estimates turn positive. In the bivariate regression, the coefficient stands at a highly significant 0.0896 , with a t -values of 7.43 . Bivariate regression shows the IVOL coefficient turning significantly positive at 0.0896 (t -values = 7.43) after controlling for MAX, which is comparable to the reversal documented by BCW who found positive coefficients around 0.385 in similar specifications. This pattern mirrors Annaert *et al.* (2013), who document a significant MAX coefficient of -0.0665 ($t = -2.54$) and a positive IVOL coefficient of 0.0924 ($t = 0.96$) in their bivariate regression.

However, in the multivariate regression incorporating the complete set of control variables (including MAX), the IVOL coefficient is negative but lacks statistical significance. Notably, the MAX coefficient remains negative and significant at -0.127 (t -values of -23.74), indicating the robustness of the MAX effect even with IVOL included in the regression model. This coefficient demonstrates stronger significance than equity market findings, with BCW reporting -0.0886 ($t = -6.67$) and Walkshäusl (2014) documenting -0.090 ($t = 6.32$) for equal-weighted and -0.068 ($t = 4.16$) for value-weighted returns.

Furthermore, Table 7 presents the results of the bivariate sorts. Initially, I construct five portfolios ranked on IVOL and then further sort for MAX within these portfolios, resulting in a 5×5 matrix. The key statistic here is the mean return differential concerning low and high MAX portfolios, which ranges from 0.0568 – 0.0703 with a t -values of well above 15 in all cases. My findings regarding hypotheses $H2a$ and $H3a$ align consistently across different equity markets and the CC market. After controlling for MAX, it is observed that the IVOL puzzle disappears or is linked with a positive premium. In other markets, Zhong and Gray (2016) report differences of 0.52 and 1.83 for equal-weighted and value-weighted portfolios in the Australian market, while BCW finds a difference of 0.91 and 0.35 for equal-weighted and value-weighted portfolios in the US equity market.

4.4 Skewness and MAX

Several studies have investigated the association between skewness and expected returns. Early research, including Harvey and Siddique (2000), Kraus and Litzenberger (1976) and Smith (2007), has underscored the importance of COSKEW by incorporating it into asset pricing models. Conversely, several studies argue that investors often hold under-diversified

Table 6. Cross-sectional regressions on IVOL, MAX and controls: This table displays the average coefficient estimates from the cross-sectional regressions on the IVOL over the previous hour, the MAX over the previous hour and other control variables

	Mod1	Mod2	Mod3	Mod4	Mod5	Mod7	Mod8	Mod9
IVOL	-0.1625*** (-22.08)	-0.1722*** (-23.49)	-0.1652*** (-22.58)	-0.1608*** (-21.78)	-0.1543*** (-19.87)	-0.1551*** (-19.56)	0.0896*** (7.43)	-0.0049 (-0.39)
Beta		0.0164*** (9.59)	0.0134*** (7.45)	0.0132*** (6.94)	0.0127*** (6.63)	0.0122*** (6.39)		0.0168*** (8.53)
Mom			-0.0693*** (-21.31)	-0.0909*** (-25.63)	-0.0837*** (-23.68)	-0.0812*** (-22.89)		-0.0718*** (-20.37)
Reversal				-0.1823*** (-35.04)	-0.1741*** (-34.31)	-0.1711*** (-33.74)		-0.1604*** (-30.68)
Illiq					-0.1661 (-1.21)	-0.1674 (-1.12)		-0.0048 (-0.03)
Log vol						0.0002 (0.61)		0.0004 (1.32)
MAX							-0.1270*** (-23.74)	-0.0817*** (-14.78)
Adj. R Sq.	4.322	7.382	9.672	12.067	13.116	13.326	6.868	14.855

Note(s): Control variables include market beta, momentum (mom), illiquidity (illiq), short-term reversal (reversal) and log volume. Beta and log volume are scaled by 10². Mod1 is the univariate regression on IVOL. Mod2 to Mod7 are regression models inclusive of various control variables. Mod8 includes IVOL and MAX, whereas Mod9 includes IVOL, MAX and all the control variables. Each row presents the time-series averages of the cross-sectional regression slope coefficients and their associated Newey and West (1987) adjusted t-statistic. Adjusted R² is also provided

Source(s): Authors' own work

Table 7. Returns on portfolios of stocks sorted by MAX after controlling for IVOL: Double-sorted equal-weighted (EW) five portfolios are formed from January 2020 to December 2022. Panels A, B and C sort CCs based on maximum daily returns MAX (1), MAX (2) and MAX (3), respectively, after adjusting for IVOL.

	Low	MAX2	MAX3	MAX4	High	Low-High
<i>Panel A: MAX (1)</i>						
Excess return	0.0312*** (6.79)	0.0200*** (3.32)	0.0129** (2.15)	0.0096 (1.56)	-0.0257*** (-4.21)	0.0568*** (17.13)
CAPM alpha	0.03100*** (6.67)	0.0200*** (3.41)	0.0130** (2.23)	0.0100 (1.60)	-0.0250*** (-4.118)	0.0560*** (15.91)
<i>Panel B: MAX (2)</i>						
Excess return	0.0365*** (8.02)	0.0211*** (3.57)	0.0116* (1.96)	0.0091 (1.49)	-0.0288*** (-4.72)	0.0653*** (18.92)
CAPM alpha	0.0360*** (7.82)	0.0210*** (3.65)	0.0120** (1.97)	0.0090 (1.54)	-0.0290*** (-4.647)	0.0650*** (17.95)
<i>Panel C: MAX (3)</i>						
Excess return	0.0401*** (8.81)	0.0218*** (3.75)	0.0126** (2.09)	0.0055 (0.91)	-0.0302*** (-4.94)	0.0703*** (19.99)
CAPM alpha	0.0400*** (8.50)	0.0220*** (3.78)	0.0130** (2.13)	0.0060 (0.94)	-0.0300*** (-4.885)	0.0700*** (18.98)
<i>Panel A: MAX (1)</i>						
Excess return	0.0312*** (6.79)	0.0200*** (3.32)	0.0129** (2.15)	0.0096 (1.56)	-0.0257*** (-4.21)	0.0568*** (17.13)
CAPM alpha	0.03100*** (6.67)	0.0200*** (3.41)	0.0130** (2.23)	0.0100 (1.60)	-0.0250*** (-4.118)	0.0560*** (15.91)
<i>Panel B: MAX (2)</i>						
Excess return	0.0365*** (8.02)	0.0211*** (3.57)	0.0116* (1.96)	0.0091 (1.49)	-0.0288*** (-4.72)	0.0653*** (18.92)
CAPM alpha	0.0360*** (7.82)	0.0210*** (3.65)	0.0120** (1.97)	0.0090 (1.54)	-0.0290*** (-4.647)	0.0650*** (17.95)
<i>Panel C: MAX (3)</i>						
Excess return	0.0401*** (8.81)	0.0218*** (3.75)	0.0126** (2.09)	0.0055 (0.91)	-0.0302*** (-4.94)	0.0703*** (19.99)
CAPM alpha	0.0400*** (8.50)	0.0220*** (3.78)	0.0130** (2.13)	0.0060 (0.94)	-0.0300*** (-4.885)	0.0700*** (18.98)
Note(s): Initially, I sorted the CCs into five portfolios using the control variable (IVOL). Subsequently, within each portfolio, I sort CCs into five portfolios based on the variable of interest (MAX). The rows in the table report average returns across the five control portfolios to produce five portfolios with dispersion in the MAX but with similar levels of the IVOL. "Excess return" indicates the difference in average returns between portfolio 5 and 1, while "CAPM alpha" is the difference in CAPM alphas between portfolio 1 and 5. Newey and West (1987) adjusted t-statistics are reported in parentheses						
Source(s): Authors' own work						

portfolios to exploit the SKEW, thus suggesting that both total skewness and ISKEW are priced components (Conine and Tamarkin, 1981; Mitton and Vorkink, 2007).

The MAX measure proposed by BCW is purely idiosyncratic in nature, producing anomalous findings where high MAX stocks tend to earn low subsequent returns. However, it does not rule out the possibility that MAX is not capturing the systematic skewness of stocks. This prompts the question of whether the idiosyncratic nature of MAX makes it robust to systematic skewness of the security. Further, unlike SKEW, which lacks theoretical support in the literature, studies like Barberis and Huang (2008) and Brunnermeier *et al.* (2007) suggest that low probability extreme return states rather than skewness directly drive results, gaining support from CPT (Tversky and Kahneman, 1992).

To assess whether MAX provides similar information to SKEW, in Table 8, I explore the robustness of the association between MAX and future returns while accounting for total SKEW, ISKEW and COSKEW. Initially, I perform bivariate sorts analysis on MAX while controlling for skewness. This entails creating five portfolios based on each skewness measure (SKEW, ISKEW or COSKEW) and then sorting CCs within each skewness portfolio into five additional portfolios based on MAX. This ensures that Portfolio 1 (5) includes CCs with the lowest (highest) MAX.

Post-adjusting for total SKEW, the mean return (CAPM alpha) gap between low and high MAX portfolios is 0.0503% (0.050%) with a Newey-West *t*-statistic of 13.16 (12.83). This suggests that total skewness does not explain the return disparity between high and low MAX CCs. Similar findings emerge from the seventh and twelfth rows, representing bivariate portfolios constructed on MAX after adjusting for ISKEW and COSKEW, respectively.

Table 9 shows cross-sectional FMB regression results, including skewness, ISKEW and COSKEW as control variables. It presents the average slope coefficients over the sample period, along with Newey-West adjusted *t*-statistics. All skewness measures are negatively associated with the expected returns in univariate regression models. These results mirror the previous research by Harvey and Siddique (2000), Smith (2007) and Mitton and Vorkink (2007) for systematic skewness and ISKEW. Additionally, incorporating any of the skewness variants has only a minimal impact on MAX, as indicated by consistently negative average coefficients across all specifications, with the smallest magnitude being -0.081 . Therefore, it can be concluded that the MAX effect is not absorbed by existing measures of skewness, indicating the unique information content of MAX.

4.5 Robustness checks

4.5.1 Alternate MAX measures. I explore MAX measures, including MAX 2 (average of two highest MAX) and MAX 3 (average of three highest MAX), and reanalyze Table 4. Results indicate that transitioning from MAX to MAX 3 not only improves the MAX coefficient but also enhances its statistical significance (Table A1 in Appendix 2).

4.5.2 Robust to longer holding periods. I extend my analysis to longer holding periods, i.e. 2, 3 and 6 h. Notably, while my main conclusions remain consistent, I observe a reduction in both the coefficient and statistical significance as I move from $t + 1$ to $t + 6$ (Table A2 in Appendix 2).

4.5.3 Bitcoin as the market factor. Given Bitcoin's dominant market position and overwhelming influence in the CC market dynamics, I exclude it from my sample and use Bitcoin return as the market risk factor instead. This adjustment does not alter my main conclusions (Table A3 in Appendix 2).

4.5.4 MAX and IVOL. Since MAX and IVOL are highly correlated (see correlation matrix), the regression results might not be very reliable. Following Annaert *et al.* (2013),

Table 8. Returns on portfolios of stocks sorted by MAX after controlling for skewness measures: Double-sorted equal-weighted (EW) five portfolios are formed from January 2020 to December 2022

	Low	MAX2	MAX3	MAX4	High	Low-High
<i>Panel A: SKEW</i>						
Excess return	0.0210*** (4.39)	0.01730*** (2.99)	0.01580*** (2.65)	0.0089 (1.43)	-0.0294*** (-4.61)	0.0503*** (13.16)
CAPM alpha	0.0210*** (4.44)	0.0170*** (3.02)	0.0160*** (2.68)	0.0090 (1.46)	-0.0290*** (-4.49)	0.0500*** (12.83)
<i>Panel B: ISKEW</i>						
Excess return	0.0232*** (4.72)	0.0185*** (3.19)	0.0158*** (2.67)	0.0133** (2.23)	-0.0231*** (-3.86)	0.0462*** (13.18)
CAPM alpha	0.0230*** (4.74)	0.0180*** (3.21)	0.0160*** (2.66)	0.0130** (2.26)	-0.0230*** (-3.74)	0.0460*** (13.52)
<i>Panel C: COSKEW</i>						
Excess return	0.0288*** (5.16)	0.0205*** (3.43)	0.0170*** (2.81)	0.0129** (2.22)	-0.0343*** (-5.74)	0.0631*** (16.41)
CAPM alpha	0.0290*** (5.08)	0.0200*** (3.57)	0.0170*** (2.89)	0.0130** (2.26)	-0.0340*** (-5.37)	0.0630*** (17.08)

Note(s): Panels A, B and C sort CCs based on SKEW, ISKEW and COSKEW, respectively. Initially, I sorted the CCs into five portfolios using the control variable (SKEW, ISKEW and COSKEW). Subsequently, within each portfolio, I sort CCs into five portfolios based on the variable of interest (MAX). The rows in the table report average returns across the five control portfolios to produce five portfolios with dispersion in the MAX but with similar levels of the SKEW. “Excess return” indicates the difference in average returns between portfolio 5 and 1, while “CAPM alpha” is the difference in CAPM alphas between portfolio 5 and 1. Newey and West (1987) adjusted t-statistics are reported in parentheses

Source(s): Authors’ own work

Table 9. Cross-sectional regressions on measures of skewness, MAX and controls: This table displays the average coefficient estimates from the cross-sectional regressions involving different measures of skewness over the previous hour (SKEW, ISKEW and COSKEW), the maximum return over the previous hour (MAX) and other control variables

	Mod1	Mod2	Mod3	Mod4	Mod5	Mod6
MAX				-0.0833*** (-21.11)	-0.0810*** (-22.33)	-0.0818*** (-22.29)
Beta				0.0162*** (8.01)	0.0171*** (8.67)	0.0173*** (8.38)
Mom				-0.0724*** (-20.32)	-0.0711*** (-19.98)	-0.0735*** (-19.44)
Reversal				-0.1591*** (-30.28)	-0.1599*** (-31.07)	-0.1616*** (-30.22)
Illiq				-0.0050 (-0.03)	-0.0461 (-0.31)	0.0112 (0.08)
Log vol				0.0040 (1.44)	0.0042 (1.47)	0.0050* (1.79)
SKEW	-0.0280*** (-19.19)			0.0001 (-0.01)		
ISKEW		-0.0235*** (-20.01)			-0.0041*** (-3.51)	
COSKEW			-0.0002 (-0.75)			-0.0005*** (-2.16)
Adj. R Sq.	0.588	0.208	2.186	13.846	13.452	14.86

Note(s): Control variables include market beta (scaled by 10^3), momentum (mom), illiquidity (Illiq), short-term reversal (reversal), skewness (SKEW, scaled by 10^2), ISKEW (scaled by 10^3), COSKEW (scaled by 10^3) and log volume (scaled by 10^2). Mod1 to Mod3 represent univariate regressions on SKEW, ISKEW and COSKEW, respectively. Mod4 to Mod6 incorporate various control variables, with each skewness measure being controlled individually. Each row presents the time-series averages of the cross-sectional regression slope coefficients and their associated Newey and West (1987) adjusted t-statistic. Adjusted R^2 is also provided (%)

Source(s): Authors' own work

I take the lag of IVOL and rerun the regression models to address this concern. They advocate for this approach for three reasons: the correlation between lagged IVOL and MAX is lower than that between IVOL and MAX, lagged IVOL is highly persistent like MAX, and the IVOL puzzle persists even when lagged IVOL is considered in the model. This does not alter the main conclusions. Further, it is observed that MAX continues to demonstrate significant negative predictive power for returns, even when considering only the portion of MAX independent of IVOL (Table A4 in Appendix 2).

5. Implications, limitations and future research

This study yields several significant implications on the theoretical and practical front. First, the study's findings align with the literature contending that loss-averse investors tend to maintain an under-diversified portfolio, thus exhibiting a preference for asset features resembling lotteries. Furthermore, the analysis disentangles the MAX effect from a closely related proxy, IVOL, revealing a consistent MAX effect even when IVOL is considered. This signifies that the MAX effect represents a distinct phenomenon, while the previously documented link between IVOL and future returns, as highlighted by [Ang et al. \(2006\)](#), loses significance. Additionally, the study unveils a negative and significant relationship between subsequent CC returns and measures of skewness (total, ISKEW and COSKEW). The analysis demonstrates that the MAX effect cannot be solely attributed to skewness measures and possesses unique informational content. Thus, neither IVOL nor the different skewness variants weaken the MAX effect.

Academically, an implication of these empirical findings is that the evidence of the MAX effect supported in asset pricing literature to explain returns in traditional markets also works for this newly emerging asset class. On the practical front, the findings offer several actionable insights. For investors seeking to invest in CC markets, the findings indicate which risk factors were relevant for investors in asset pricing and help implement profitable trading strategies. Moreover, given the massive amount of professional money involved in the CC market, there are also practical implications. Findings show that taking long (short) positions on low (high) MAX CCs presents highly profitable opportunities for professional money managers [\[12\]](#). For regulators, a better understanding of these behavioral patterns can inform the development of more effective market surveillance and investor protection policies.

Several promising avenues for future research emerge from the study's findings. First, researchers could investigate the potential moderating effects of other behavioral factors, such as market uncertainty, investor attention and sentiment, on the MAX-return relationship. Second, the seasonality of lottery preferences in CC markets is also an interesting theme. While the calendar effects examined in the financial markets literature, including the CC ([Kaiser, 2019](#); [Kinateder and Papavassiliou, 2021](#); [Long et al., 2020](#)), typically manifest over longer horizons (daily, weekly or monthly patterns), subtle calendar effects might still exist at the intraday level, even if they are not as pronounced as traditional calendar effects. Thus, future studies can explore the interactions between intraday trading patterns and lottery preferences in CC markets.

6. Conclusion

Noting the increasing evidence of the speculative nature of the CCs, characterized by heightened non-linearities, tail risk and volatility, this study attempts to test a distinct speculative behavior among CC investors. Using a set of 100 largest and most liquid CCs, the study reinforces the MAX effect in the CC market. This effect manifests as a notably negative correlation between the MAX observed in the previous hour and future CC returns,

even after adjusting for established return predictors like beta, momentum, short-term reversal, illiquidity, IVOL and skewness.

The study design is analogous to seminal literature investigating the lottery-like demands in equity, other traditional asset classes and CCs. However, it diverges from existing literature by using high-frequency data to delve deeper into speculative demands. This shift in methodology is encouraged by the increasing prevalence of algorithmic trading in financial markets, particularly evident in CCs, as well as the characteristics of short memory processes exhibited by CCs, as suggested by recent literature (Grobys *et al.*, 2020). Consequently, the intraday data might offer a more comprehensive view of market dynamics. This enables sentiment shifts and order book dynamics to be captured more promptly, giving insights into the intricacies of CC investor behavior.

On the theoretical front, the literature indicates that investors often deviate from rational equilibrium and Expected Utility Theory, with considerable attention given to their preferences for lottery-like investments. The findings of my study are grounded in the very popular CPT due to Tversky and Kahneman (1992), which emphasizes investors' tendency to overweight outcomes with small probabilities, reflecting their gambling or lottery preferences. The theory further finds support in Brunnermeier *et al.*'s (2007) optimal expectation framework. In conclusion, integrating psychological and behavioral factors can also assist policymakers in effectively regulating these emerging economically significant markets.

Notes

1. I use coskewness and systematic skewness interchangeably.
2. Similar evidence is found in the Hong market (Chan & Chui, 2016), the South Korean equity market (Nartea *et al.*, 2014), the Indian equity market (Ali, Hasan, *et al.*, 2020), the Singapore equity market (Ali, Rahman, *et al.*, 2020), the Australian stock market (Zhong & Gray, 2016), and in Turkish market (Ali, Ahmed, *et al.*, 2020).
3. Binance stands as a leading CC exchange worldwide, with a daily trading volume of \$25bn. <https://coinmarketcap.com/exchanges/binance/>
4. The CC is known for its rapid evolution. Thus, relying on more recent data period (2020, 2021 and 2022) would ensure that findings are relevant to contemporary market conditions and recent trends in the market. Further, the use of minute by minute frequency ensures sufficient sample size to establish robustness of these findings.
5. Tether is a popular unit symbol used in exchanges, and it is 1:1 pegged to the USD. Guo *et al.* (2022) also use the USDT pairing for their analysis.
6. As a robustness check, I excluded Bitcoin from my sample and treated Bitcoin returns as the market return. Detailed discussion in Section 4.
7. I apply winsorization to all variables at the 1st and 99th percentiles. This ensures that the higher payoff on low MAX cannot be attributed to a few extreme observations at the tail. In the untabulated results, my analysis on raw data yields consistent findings, corroborating the robustness of my results.
8. Despite such a high correlation, the VIF of less than 2 can be attributed to my large sample size, which could stabilize the estimation of regression coefficients even when the correlation is high. However, as a robustness measure, following Annaert *et al.* (2013), I incorporate the lag of IVOL into my regression models. Despite this adjustment, the findings remain consistent. Moreover, MAX continues to demonstrate significant negative predictive power for returns, even when considering only the portion of MAX independent of IVOL (orthogonalized MAX).

9. Here, I focus on the raw average return rather than the average return adjusted for the risk-free rate, as the risk-free rate at intraday frequency is not significant. Additionally, the CAPM alpha is derived from the regression analysis on market returns.
10. The coefficient for MAX is -0.083 , with a mean cross-sectional standard deviation of 0.0052 . Therefore, the measure of the effect amounts to -0.043% ($-0.083 * 0.0052$).
11. Grobys & Sapkota (2019) and Shen *et al.* (2020) have documented a short-term reversal phenomenon in the cross-section of CCs.
12. Short-selling rules in the CC market may differ from traditional assets due to certain restrictions (Corbet *et al.*, 2018; Grobys & Junttila, 2021). Consequently, a long-only strategy could be the sole alternative.

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Further reading

Simkowitz, M.A. and Beedles, W.L. (1978), "Diversification in a Three-Moment world", *The Journal of Financial and Quantitative Analysis*, Vol. 13 No. 5, pp. 927-941, doi: [10.2307/2330635](https://doi.org/10.2307/2330635).

Appendix 1. Variable construction

The central variables of the study are defined as follows. One key variable of interest MAX_{ij} for CC i is defined as the maximum 5-min return in the hour $t-1$. I calculate $MAX(n)_{ij}$ as the mean of n maximum 5-min returns for CC i during the hour $t-1$ (where $n = 1, 2, 3$):

$$MAX(n) = \sum_{i=1}^n \text{maximum return} \quad (1)$$

Momentum (MOM) is the cumulative return for CC i for 15 min from $t-3$ to $t-2$ (Jegadeesh and Titman, 1993; Y. Liu and Tsyvinski, 2021). The return of the most recent 5 min or the lagged return is defined as a short-term reversal (Jegadeesh and Titman, 1993). I employ Amihud's illiquidity (ILLIQ) measure, calculated as the absolute return divided by its dollar trading volume (Amihud, 2002). Trading volume is expressed as the natural logarithm of CC i 's average trading volume in hour $t-1$. Skewness represents the overall skewness, computed as the natural measure of the third central moment using returns over the previous hour. I take the equal-weighted average of all the CCs in the sample as the market return. I decompose the skewness into idiosyncratic and co-skewness components by estimating the following regression (Bali et al., 2011d; Harvey and Siddique, 2000b):

$$R_{i,t} = \alpha_i + \beta_i(R_{m,t}) + \gamma_i(R_{m,t})^2 + \epsilon_{i,t} \quad (2)$$

where, $R_{i,t}$ is the return on CC i in hour t , $R_{m,t}$ is the market return in hour t . Idiosyncratic skewness of CC i in hour t is the skewness of residuals $\epsilon_{i,t}$. Systematic skewness of CC i in hour t is the slope (γ_i) of $(R_{m,t})^2$. Following (Dimson, 1979; Scholes and Williams, 1977), I use the lag and lead of the market return along with the current period market when calculating beta:

$$R_{i,t} = \alpha_i + \beta_{1,i}(R_{m,t-1}) + \beta_{2,i}(R_{m,t}) + \beta_{3,i}(R_{m,t+1}) + \epsilon_{i,t} \quad (3)$$

I estimate equation (3) for each CC using 5-min returns within an hour. Subsequently, the market beta for each CC i in hour t is defined as:

$$\widehat{\beta} = \widehat{\beta}_{1,i} + \widehat{\beta}_{2,i} + \widehat{\beta}_{3,i} \quad (4)$$

The total volatility of CC i in hour t is the standard deviation (SD) of daily returns within hour t :

$$TVOL_{i,t} = \sqrt{\text{var}}(R_{i,t}) \quad (5)$$

Idiosyncratic volatility is calculated using the Fama-French market model. From the following equation, idiosyncratic volatility is the standard deviation of $\epsilon_{i,t}$ or $[IVOL_{i,t} = \sqrt{\text{var}}(\epsilon_{i,t})]$:

$$R_{i,t} = \alpha_i + \beta_i(R_{m,t}) + \epsilon_{i,t} \quad (6)$$

Table A1. Cross-sectional regressions MAX (N) and controls; for each hour from 2020-01 to 01:00-2022-11-18:01, I conduct hourly cross-sectional regression of the return in that hour on subsets of lagged predictor variables, including MAX in the previous hour and six control variables

	Mod1	Mod2	Mod3	Mod4	Mod5	Mod6
<i>Panel A: MAX 2</i>						
MAX	-0.1271*** (-28.529)	-0.1459*** (-32.313)	-0.137*** (-29.772)	-0.1217*** (-26.435)	-0.1169*** (-24.374)	-0.1175*** (-24.248)
Beta		0.0265*** (14.83)	0.0219*** (11.81)	0.0211*** (10.64)	0.0196*** (9.90)	0.0186*** (9.44)
Mom			-0.0569*** (-17.622)	-0.078*** (-22.121)	-0.0725*** (-20.428)	-0.0703*** (-19.727)
Reversal				-0.1667*** (-31.568)	-0.1602*** (-31.122)	-0.1577*** (-30.642)
Illiq					0.1045 (0.77)	0.1614 (1.09)
Log vol						0.0007*** (2.67)
Adj. RSq.	5.136	7.826	10.023	12.311	13.334	13.546
<i>Panel B: MAX 3</i>						
MAX	-0.1588*** (-29.08)	-0.1823*** (-32.86)	-0.1711*** (-30.23)	-0.1496*** (-26.32)	-0.144*** (-24.37)	-0.145*** (-24.18)
Beta		0.0271*** (15.19)	0.0222*** (12.00)	0.0212*** (10.68)	0.0196*** (9.83)	0.0185*** (9.35)
Mom			-0.0554*** (-17.08)	-0.0766*** (-21.67)	-0.0716*** (-20.13)	-0.0695*** (-19.47)
Reversal				-0.1644*** (-31.30)	-0.1582*** (-30.75)	-0.1556*** (-30.25)
Illiq					0.1524 (1.12)	0.2307 (1.55)
Log vol						0.0009*** (3.14)
Adj. RSq.	5.308	7.975	10.153	12.426	13.451	13.661

Note(s): The set of control variables includes market beta (BETA), momentum (MOM), illiquidity (Illiq), short-term reversal (reversal), skewness (SKEW) and log volume. Beta, skewness and log volume are scaled by 10². This procedure yields a time series of slope coefficients. In each row, the table presents the time-series averages of the cross-sectional regression slope coefficients and their associated Newey-West adjusted t-statistics (in parentheses)

Source(s): Authors' own work

Table A2. Longer holding periods: For each hour from 2020–01 to 01:00–2022–11–18:01, I conduct hourly cross-sectional regression of the return not only on $t + 1$ h but also $t + 2$, $t + 3$ and $t + 6$ on subsets of lagged (at t) predictor variables, including MAX in the previous hour and six control variables

	MAX	Beta	Mom	Reversal	Illiq	Log vol	Adj R Sq.
<i>Panel A: $t + 1$</i>							
MAX univariate	-0.0928*** (-27.79)						4.733
MAX with controls	-0.083*** (-23.15)	0.0173*** (8.82)	-0.0718*** (-20.23)	-0.1602*** (-31.02)	-0.0669 (-0.46)	0.0004 (1.46)	13.317
<i>Panel B: $t + 2$</i>							
MAX univariate	-0.0483*** (-15.12)						3.996
MAX with controls	-0.0459*** (-13.11)	0.0051*** (2.73)	-0.023*** (-7.19)	-0.0245*** (-5.51)	-0.2393* (-1.68)	-0.0001 (-0.07)	11.119
<i>Panel C: $t + 3$</i>							
MAX univariate	-0.0343*** (-10.43)						3.767
MAX with controls	-0.0319*** (-9.33)	0.0018 (0.97)	-0.0059* (-1.80)	-0.0076* (-1.65)	-0.4401*** (-3.17)	-0.0001 (-1.77)	10.579
<i>Panel D: $t + 6$</i>							
MAX univariate	-0.0174*** (-5.70)						3.411
MAX with controls	-0.021*** (-6.28)	0.0021 (1.13)	-0.0015 (-0.49)	0.0028 (0.65)	0.0447 (0.32)	-0.0002 (-0.65)	9.507

Note(s): The set of control variables includes market beta (BETA), momentum (MOM), illiquidity (illiq), short-term reversal (reversal), skewness (SKEW) and log volume. Beta, skewness and log volume are scaled by 10^2 . This procedure yields a time series of slope coefficients. In each row, the table presents the time-series averages of the cross-sectional regression slope coefficients and their associated Newey-West adjusted t-statistics (in parentheses)

Source(s): Authors' own work

Table A3. Excluding bitcoin from the sample: In panel A, for each hour from 2020-01 to 01:00-2022-11-18:01, I conduct hourly cross-sectional regression of the return in that hour on subsets of lagged predictor variables, including MAX in the previous hour and six control variables

	Mod1	Mod2	Mod3	Mod4	Mod5	Mod6
<i>Panel A: Cross-sectional regressions MAX and controls</i>						
MAX	-0.0935*** (-28.019)	-0.1033*** (-30.965)	-0.0973*** (-28.824)	-0.0882*** (-26.071)	-0.0829*** (-23.739)	-0.0823*** (-23.375)
Beta		0.0165*** (12.00)	0.0136*** (9.48)	0.014*** (9.11)	0.0128*** (8.32)	0.0122*** (8.04)
Mom			-0.061*** (-18.461)	-0.0809*** (-22.671)	-0.075*** (-21.006)	-0.0729*** (-20.36)
Reversal				-0.1686*** (-32.527)	-0.1616*** (-32.093)	-0.1589*** (-31.469)
Illiq					-0.0806 (-0.597)	-0.0778 (-0.535)
Log vol						0.0004 (1.28)
Adj. R Sq.	4.703	7.417	9.672	11.984	13.014	13.246
<i>Panel B: Cross-sectional regressions IVOL and controls</i>						
IVOL	-0.1531*** (-20.67)	-0.1627*** (-22.064)	-0.157*** (-20.978)	-0.1501*** (-19.173)	-0.1516*** (-18.983)	0.1173*** -9.721
Beta		0.0106*** (7.83)	0.0085*** (6.10)	0.0085*** (5.68)	0.0084*** (5.58)	-0.1389*** (-25.595)
Mom			-0.0711*** (-21.271)	-0.0847*** (-23.53)	-0.0826*** (-22.853)	0.0116*** (7.60)
Reversal			-0.0914*** (-25.308)	-0.0847*** (-34.585)	-0.169*** (-34.077)	-0.073*** (-20.177)
Illiq			-0.1795*** (-35.109)	-0.1721*** (-34.585)	-0.1789 (-1.216)	-0.1599*** (-31.438)
Log vol				-0.1899 (-1.399)	0.0001 (0.44)	-0.0611 (-0.407)
MAX						0.0024 (0.85)
Adj. R Sq.	4.575	7.426	12.088	13.121	13.329	-0.0831*** (-14.876)
					6.84	14.782
<i>Panel C: Cross-sectional regressions skewness and controls</i>						
MAX						Mod6
Beta						-0.0798*** (-22.045)
Mom						0.0105*** (6.57)
Reversal						-0.0756*** (-19.893)
Illiq						-0.1574*** (-30.184)
Log vol						-0.0527 (-0.363)
SKEW	-0.0281*** (-19.095)					0.0041 (1.41)
ISKEW		-0.0252*** (-21.59)				
COSKEW			0.0001 (0.46)			
Adj. R Sq.	0.587	0.26	2.344	13.779	13.424	-0.0002 (-1.081)
						14.792

Note(s): The sample excludes the bitcoin, and the bitcoin return is taken as the market return. The set of control variables includes market beta (BETA), momentum (MOM), illiquidity (illiq), short-term reversal (reversal), skewness (SKEW) and log volume. Beta, skewness and log volume are scaled by 10^4 . This procedure yields a time series of slope coefficients. In each row, the table presents the time-series averages of the cross-sectional regression slope coefficients and their associated Newey-West adjusted t-statistics (in parentheses). Panel A represents the cross-sectional regressions MAX and controls. Panel B represents the cross-sectional regressions on IVOL, MAX and controls, and panel C represents the cross-sectional regressions on measures of skewness, MAX and controls

Source(s): Authors' own work

Table A4. Cross-sectional regressions on IVOL, MAX and controls: Panel A of the table displays the average coefficient estimates from the cross-sectional regressions on the lagged idiosyncratic volatility ($IVOL_{t-1}$), the maximum return (MAX) and other control variables

	Mod1	Mod2	Mod3	Mod4	Mod5	Mod7	Mod8	Mod9
<i>Panel A: Lagged IVOL, FMB cross-section regressions</i>								
IVOL	-0.1065*** (-14.646)	-0.1112*** (-15.48)	-0.1108*** (-15.45)	-0.1107*** (-15.26)	-0.0981*** (-13.11)	-0.097*** (-12.79)	0.0159** (2.02)	-0.0265*** (-3.29)
Beta	0.0135*** (8.03)	0.0103*** (5.92)	0.01*** (5.44)	0.0099*** (5.28)	0.0097*** (5.17)	0.0097*** (5.17)	-0.0967*** (-26.75)	0.0164*** (8.40)
Mom			-0.0721*** (-22.01)	-0.0933*** (-26.19)	-0.0836*** (-23.64)	-0.081*** (-22.75)		-0.0734*** (-20.42)
Reversal				-0.1852*** (-35.08)	-0.1749*** (-34.24)	-0.1716*** (-33.5)		-0.1618*** (-31.41)
Illiqu					-0.7825*** (-6.013)	-0.9258*** (-6.575)		0.0072 (0.05)
Log vol						-9e-04*** (-3.44)		0.0005 (1.70)
MAX								-0.076*** (-20.15)
Adj. R Sq.	3.809	6.967	9.383	11.863	13.088	13.363	7.105	15.186
<i>Panel B: Orthogonalized IVOL and MAX</i>								
IVOL	0.0896*** (7.43)							
Max	-0.127*** (-23.734)	-0.1007*** (-29.628)		-0.1776*** (-22.729)	-0.0049 (-0.388)	-0.0817*** (-14.782)	-0.0832*** (-22.887)	-0.1769*** (-21.026)
MAX orth		0.0896*** (7.43)					-0.0049 (-0.388)	
Beta				-0.127*** (-23.734)	0.0168*** (8.53)	0.0168*** (8.53)	0.0168*** (8.53)	-0.0817*** (-14.782)
Mom					-0.0717*** (-20.349)	-0.0717*** (-20.349)	-0.0717*** (-20.349)	0.0168*** (8.53)
Reversal					-0.1604*** (-30.685)	-0.1604*** (-30.685)	-0.1604*** (-30.685)	-0.0717*** (-20.349)
Illiqu					-0.0047 (-0.031)	-0.0047 (-0.031)	-0.0047 (-0.031)	-0.1604*** (-30.685)
Log vol					0.0001 (1.33)	0.0001 (1.33)	0.0001 (1.33)	-0.0050 (-0.031)
Adj. R Sq.	6.868	6.868	6.868	6.868	14.855	14.855	0.0001 (1.33)	0.0001 (1.33)
								14.855
Notes(s): Control variables include market beta, momentum (mom), illiquidity (illiqu), short-term reversal (reversal), skewness (SKEW) and log volume. Beta, skewness and log volume are scaled by 10 ² . Mod1 is the univariate regression on IVOL. Mod2 to Mod7 are regression models inclusive of various control variables. Mod8 includes IVOL and MAX, whereas Mod9 includes IVOL, MAX and all the control variables. Each row presents the time-series averages of the cross-sectional regression slope coefficients and their associated Newey-West (1987) adjusted t-statistic. Adjusted R ² is also provided. In panel B, I reassess the return predictability of MAX and IVOL, given their high cross-sectional correlation of 78.6% (see Table 1 of the main paper), and present the Fama-MacBeth regression estimates. The first model indicates that when MAX is incorporated into the model, the IVOL puzzle is reversed, while the MAX effect remains robust. In Mod2, I introduce IVOL orthogonalized to MAX as an additional explanatory variable alongside MAX. Orthogonalized IVOL denotes a CC's residual from cross-sectional regressions of IVOL on MAX, signifying the portion of IVOL independent of MAX. Similarly, in Mod3, IVOL and MAX orthogonalized to IVOL are employed as regressors: while the notably negative IVOL coefficient confirms the IVOL puzzle, even the portion of MAX orthogonal to IVOL significantly predicts subsequent returns with a negative sign. Mod4 to Mod6 reaffirm these findings, indicating that the conclusions remain consistent when including the standard set of control variables as independent variables. Therefore, Table A4 corroborates the main paper's argument that MAX surpasses IVOL in predicting subsequent returns in the sample								
Source(s): Authors' own work								

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